A Diagrammatic Proof of the Haldane Gap via Spacetime Tensor Networks

(Dated: October 8, 2026)

Following the recent machine-assisted resolution of the Haldane conjecture for periodic and boundary-field spin-1 Heisenberg chains by OpenAI [1, 2], we show that the analytical derivations can be recast as elementary tensor-network contractions. Formulating the system as an exact continuous-time thermal matrix product operator exposes the underlying spacetime geometry. The step from a finite-scale estimate to a uniform gap then reduces to a contractive scale-doubling recurrence on planar diagrams, under three pages long and formalized in Lean 4 with the two numerical inputs as its only unproved hypotheses. The finite-scale estimate itself is taken unchanged from Ref. [1], whose certificate uses spectral filters and character projections. This illustrates how diagrammatic representations can clarify and compress machine-generated proofs in quantum many-body physics.

We consider the antiferromagnetic spin-1 Heisenberg chain with periodic boundary conditions $\boldsymbol{S}_{L+1} = \boldsymbol{S}_1$ and $J = k_B = 1$:

$$H_L = \sum_{j=1}^{L} \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}, \qquad Z_L(\beta) = \operatorname{Tr} e^{-\beta H_L}. \tag{1}$$

We take even $L \geq 4$ so the transfer trace is an even power. Let $E_0(L) \leq E_1(L) \leq \cdots$ include multiplicities, with gap $\Delta_L = E_1(L) - E_0(L)$ ( ) and Gibbs weights $w_m = e^{-\beta E_m(L)}/Z_L(\beta)$:

$$\frac{w_1}{w_0} = e^{-\beta\Delta_L} \leq e^{-c\beta} \quad \Longrightarrow \quad \Delta_L \geq c. \tag{2}$$

The ratio decays in β like ; the task is to make its rate c independent of L:

Theorem 1 (Periodic gap [1])**.** *For every even $L \geq 2304$, the chain (1) has a unique ground state and $\Delta_L > (\log 20)/784 \simeq 0.003821$.*

Theorem 1 was established by OpenAI [1]. After showing how a thermal purity near one forces a gap, our proof has three stages: (i) an exact continuous-time thermal MPO yields a self-adjoint spatial transfer operator; (ii) spacetime contraction identities couple spatial and thermal purities so joint scale doubling quadratically suppresses both defects (the *spacetime bootstrap*); and (iii) interpolating between successive doubled lengths covers all even $L \geq 2304$.

Why tensor-network diagrams? In Eq. (1), L counts sites in a sum whereas β sits in an exponential. Drawing $Z_L(\beta)$ as a two-dimensional tensor network [3, 4] puts space and imaginary time on equal footing on a torus: vertical double legs ‖ carry the physical spin space through imaginary time (trace Tr; blue, thermal ρ), horizontal legs — couple neighboring sites in the virtual bond space (trace $\mathrm{Tr_v}$; red, spatial σ), joined legs are contracted, and closed loops are traces. Reading the same network along either axis turns the proof into planar diagram identities; Appendix A caps the horizontal ends with boundary vectors for an open chain.

From purity to gap.— For a positive trace-one operator with eigenvalues $p_0 \geq p_1 \geq \cdots$, the purity obeys

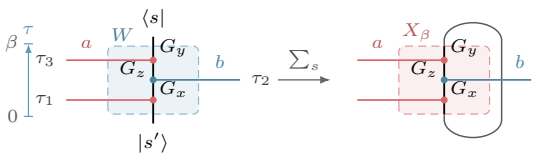
$$P = \sum_i p_i^2 \leq p_0 \sum_i p_i = p_0. \tag{3}$$

Thus the *purity defect* $1 - P$ bounds $1 - p_0 \leq 1 - P$: a purity near one forces a spectrum like with a single dominant bar. For the Gibbs state, $P > 1/2$ implies a unique ground state, with $1 - w_0 \leq 1 - P$ and $e^{-\beta\Delta_L} = w_1/w_0 \leq (1-P)/P$. To reach Eq. (2) it suffices to show that $1 - P$ falls like $e^{-c\beta}$ uniformly in L. Squaring in imaginary time improves purity at fixed L [Eq. (10)], but growing L adds thermal excitations; advancing L and β together requires a spatial transfer operator.

Thermal MPO.— As a warm-up, consider the classical Ising ring with spins $t_j = \pm 1$. Its Boltzmann weight $\prod_j e^{\beta t_j t_{j+1}}$ is a product of entries of the 2×2 matrix $K_{tt'} = e^{\beta t t'}$, and each spin appears in exactly two neighboring factors. Summing a spin shared by two factors is a matrix product, $\sum_{t_2} K_{t_1 t_2} K_{t_2 t_3} = (K^2)_{t_1 t_3}$, which is what a joined leg means; summing every spin multiplies K around the ring, and the closure $t_{n+1} = t_1$ turns the last sum into a trace, $Z_n = \operatorname{Tr} K^n$. Here we construct a real symmetric operator X_β on the virtual bond space that plays the role of K, with $Z_n(\beta) = \mathrm{Tr_v}\, X_\beta^n$. In the Cartesian basis $s, s' \in \{x, y, z\}$, $(S^\alpha)_{ss'} = -i\epsilon_{\alpha ss'}$ gives real operators $G_\alpha = iS^\alpha$ with entries $(G_\alpha)_{ss'} = \epsilon_{\alpha ss'}$: $G_\alpha v = v \times e_\alpha$ turns the plane perpendicular to e_α by a quarter turn and annihilates e_α (for example $G_z e_x = -e_y$ and $G_z^2 = \mathrm{diag}(-1, -1, 0)$), so $\|G_\alpha\| = 1$. Because $G_\alpha \otimes G_\alpha = i^2 S^\alpha \otimes S^\alpha$, $-H_L = \sum_{j,\alpha} G_{\alpha,j} G_{\alpha,j+1}$. Since the simplex $0 < \tau_1 < \cdots < \tau_k < \beta$ in imaginary time τ has volume $\beta^k/k!$, $e^{\beta A} = \sum_{k\geq 0} \int_{0<\tau_1<\cdots<\tau_k<\beta} d\tau_1 \cdots d\tau_k\, A^k$. For $A = -H_L$, each factor selects a bond and a label α, inserting G_α on both sites of that bond at time τ. We call such a factor an *event* (τ, α) on its bond, and the finite set of events on one bond its *bond history*; each term of the expansion is a choice of history on every bond. For the histories a and b of the left and right bonds of a site, the local MPO tensor multiplies the matrices hitting its worldline in time order ($\mathcal{T}_\tau$ placing later times to the left):

$$a \;\; s \;\; b \;\; s' \;\; = W_\beta^{ss'}(a,b) = \left\langle s \middle| \mathcal{T}_\tau \prod_{(\tau,\alpha)\in a\cup b} G_\alpha \middle| s' \right\rangle. \tag{4}$$

For $a = \{(\tau_1, x), (\tau_3, y)\}$ and $b = \{(\tau_2, z)\}$ with $\tau_1 < \tau_2 < \tau_3$, the site worldline and its physical trace $= $ are

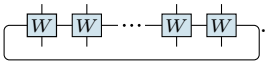


$W_\beta^{ss'}(a,b) = \langle s|G_yG_zG_x|s'\rangle$ $\quad X_\beta(a,b) = \mathrm{Tr}(G_yG_zG_x)$

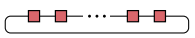
Operators on different sites commute, so once the events on every bond are fixed, the time-ordered product over the ring splits into one worldline per site, whose physical trace is $X_\beta(a,b) = \sum_s W_\beta^{ss}(a,b)$. The bond history plays the role of the Ising spin, and X_β the role of K. Summing the three labels $\alpha \in \{x,y,z\}$ and integrating ordered times defines a measure μ_β of mass $\sum_{k\ge0}(3\beta)^k/k! = e^{3\beta}$ (unit mass on the empty history). With a_j the history of bond $(j, j+1)$ and $a_0 = a_n$,

$$Z_n(\beta) = \int \prod_{j=1}^{n} d\mu_\beta(a_j) \prod_{j=1}^{n} X_\beta(a_{j-1}, a_j), \tag{5}$$

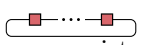
and integrating each shared history a_j is the matrix product of the Ising ring, now on $L^2(\mu_\beta)$. Contracting n tensors around the ring recovers the convergent exponential series as an exact MPO on $L^2(\mu_\beta)$ without bond truncation [3, 4]:

$e^{-\beta H_n} =$  . (6)

The left leg a is the row index of X_β and the right leg b its column index, so symmetry means that a site cannot tell its left bond from its right one. Because a and b enter Eq. (4) through $a \cup b$ and each event inserts the same real matrix G_α at both ends of its bond (*bond reflection positivity*, $-\boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1} = \sum_\alpha G_{\alpha,j} G_{\alpha,j+1}$), the spatial transfer kernel X_β is real and symmetric ($X_\beta(a,b) = X_\beta(b,a)$). The condition is not automatic: for a bond term $-h = Q \otimes P$ with $Q = \mathrm{diag}(1,1,0)$ and $P = \mathrm{diag}(1,0,0)$, a site receives P through its left bond and Q through its right one, so a history a of two events, with the empty history $\emptyset$ on the other bond, gives $X_\beta(a,\emptyset) = \mathrm{Tr}\, P^2 = 1$ from the left but $X_\beta(\emptyset,a) = \mathrm{Tr}\, Q^2 = 2$ from the right. In general, regrouping the four legs of $-h$ by site (rows $s_j s'_j$, columns $s_{j+1} s'_{j+1}$) turns $-h = \sum_\alpha G_\alpha \otimes G_\alpha$ into the 9×9 matrix $\sum_\alpha \mathrm{vec}\, G_\alpha \,\mathrm{vec}\, G_\alpha^{\mathsf{T}}$, a sum of squares; a positive semidefinite regrouped bond is the reflection positivity that the general criterion requires [5]. Since $\|G_\alpha\| = 1$ on $\mathbb{C}^3$, $|X_\beta(a,b)| \le \mathrm{Tr}\, I = 3$; with the finite mass $e^{6\beta}$ of $\mu_\beta \times \mu_\beta$, X_β is self-adjoint and Hilbert–Schmidt on $L^2(\mu_\beta)$, giving a discrete real spectrum $|\lambda_0(\beta)| \ge |\lambda_1(\beta)| \ge \cdots$ ($\sum_i \lambda_i(\beta)^2 < \infty$). Closing the ring of transfer vertices gives, for even n,

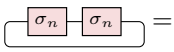
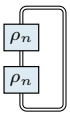
$Z_n(\beta) =$  $= \mathrm{Tr_v}\, X_\beta^n = \sum_i |\lambda_i(\beta)|^n,$ (7)

so raising the eigenvalues to the nth power concentrates the weight on λ_0:  ($n = 4$).

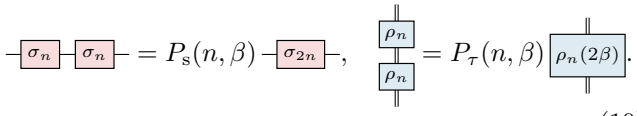
Spatial and thermal purities.— The transfer operator gives space its own density operator. Cut the ring of Eq. (7) at one bond,  : what is left is X_β^n with two open virtual legs. Cutting the torus instead along one time slice leaves $e^{-\beta H_n}$ with open physical legs, Eq. (6). Normalizing both to unit trace gives

$$\sigma_n(\beta) = \frac{\text{  }}{Z_n(\beta)} = \frac{X_\beta^n}{Z_n(\beta)}, \qquad \rho_n(\beta) = \frac{e^{-\beta H_n}}{Z_n(\beta)}. \tag{8}$$

For even n, both operators are positive with unit trace; denote their eigenvalues by $p_i = |\lambda_i(\beta)|^n / Z_n(\beta)$ and w_m, respectively. Their purities join two copies, along space for σ_n and along imaginary time for ρ_n. Drawing $Z_n(\beta)$ as a spacetime torus  of width n and height β (dashed edges identified), both are one ratio of tori:

$$\begin{aligned} P_{\mathrm{s}}(n,\beta) &= \mathrm{Tr_v}\, \sigma_n^2 = \text{  } = \frac{\mathrm{Tr_v}\, X_\beta^{2n}}{Z_n(\beta)^2} \\ &= \frac{Z_{2n}(\beta)}{Z_n(\beta)^2} = \frac{\text{}}{\text{}^2}, \\ P_\tau(n,\beta) &= \mathrm{Tr}\, \rho_n^2 = \text{  } = \frac{\mathrm{Tr}\, e^{-2\beta H_n}}{Z_n(\beta)^2} \\ &= \frac{Z_n(2\beta)}{Z_n(\beta)^2} = \frac{\text{}}{\text{}^2}. \end{aligned} \tag{9}$$

Squaring.— Squaring improves a purity: squaring σ_n doubles the length; squaring ρ_n doubles the inverse temperature. Indeed $\sigma_n^2 = X_\beta^{2n}/Z_n^2 = (Z_{2n}/Z_n^2)\,\sigma_{2n}$, and likewise for ρ_n^2:

 (10)

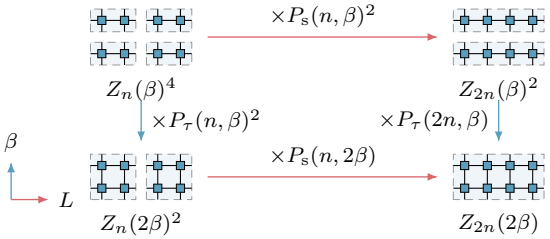
Both replace the normalized eigenvalues p_i by p_i^2/P, concentrating the spectrum:  $\mapsto$  . Since $1 - P = 2\sum_{i<k} p_i p_k$, dropping cross terms from $(\sum_{i<k} p_i p_k)^2$ bounds the new defect $1 - P'$ via $f(z) := z^2/[2(1-z)^2]$:

$$1 - P' = \frac{2\sum_{i<k} p_i^2 p_k^2}{P^2} \le \frac{(1-P)^2}{2P^2} = f(1-P). \tag{11}$$

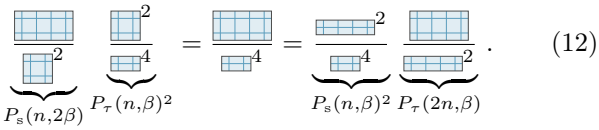
Because f increases on $[0,1)$ with $f(z) \sim z^2/2$, squaring roughly squares the purity defect $1 - P$, keeping it far below the diagonal, for $0 \le z \le .3$.

Spacetime coupling identities.— Each squaring doubles one side of the torus: space ($n \to 2n$),  $\mapsto$  , or imaginary time ($\beta \to 2\beta$),  $\mapsto$  . Because a uniform gap needs both sides to grow together, we need identities trading purity in one direction for purity in the other. Each gluing multiplies by exactly one purity, because each purity is a ratio of tori [Eq. (9)]: $Z_{2n}(\beta) = P_{\mathrm{s}}(n,\beta)\, Z_n(\beta)^2$ and $Z_n(2\beta) = P_\tau(n,\beta)\, Z_n(\beta)^2$. Starting from four tori $Z_n(\beta)^4$, gluing pairs first in space and then in time, or first in time and then in space, reaches the same doubled torus $Z_{2n}(2\beta)$ (red arrows glue in space,

blue in time, labeled by the purity factor they multiply by):



Writing each purity as its ratio of tori, Eq. (9), the two paths cancel to the same quotient:



(12)

Solving for one factor, and repeating at length $2n$, gives

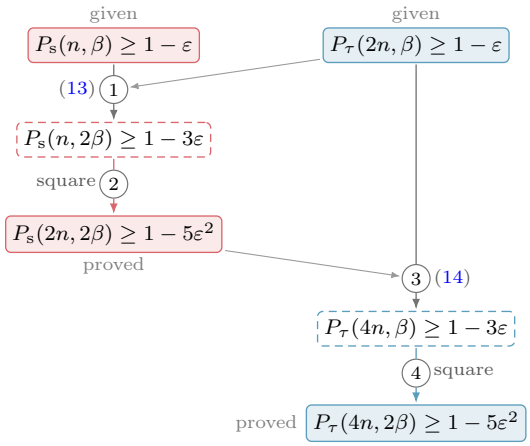
$$P_s(n,2\beta) = \frac{P_s(n,\beta)^2 P_\tau(2n,\beta)}{P_\tau(n,\beta)^2}, \tag{13}$$

$$P_\tau(4n,\beta) = \frac{P_\tau(2n,\beta)^2 P_s(2n,2\beta)}{P_s(2n,\beta)^2}. \tag{14}$$

Doubling estimate.— Alternating the two coupling identities with the two squarings doubles the torus in both directions (▫▭▭) and roughly squares the purity defect. For even n and $0 \le \varepsilon \le 1/100$,

$$\begin{aligned} &P_s(n,\beta),\ P_\tau(2n,\beta) \ge 1-\varepsilon \\ \Longrightarrow\quad &P_s(2n,2\beta),\ P_\tau(4n,2\beta) \ge 1-5\varepsilon^2. \end{aligned} \tag{15}$$

Four moves prove Eq. (15). Read the diagram below from top to bottom (red boxes bound P_s, blue P_τ; solid given or proved, dashed intermediate; circles label the identity and grey arrows supply the second input):



The proved pair is the given pair at $(2n, 2\beta)$, so Eq. (15) iterates.

Move 1 (coupling). Since $P_\tau(n,\beta) \le 1$, Eq. (13) gives $P_s(n,2\beta) \ge (1-\varepsilon)^3 \ge 1-3\varepsilon$. *Move 2* (squaring in space). For even n, $\sigma_n(2\beta)$ has nonnegative spectrum and squares to $\sigma_{2n}(2\beta)$. Since $1-(1-\varepsilon)^3 \le 3\varepsilon$ and $9/[2(1-3\varepsilon)^2] \le 5$ for $\varepsilon \le 1/100$, Eq. (11) and monotonicity of f give $P_s(2n,2\beta) \ge 1-\delta$, with

$$\delta = f\big(1-(1-\varepsilon)^3\big) = \tfrac{1}{2}\big[(1-\varepsilon)^{-3}-1\big]^2 \le 5\varepsilon^2 \le \varepsilon. \tag{16}$$

Move 3 (coupling). Equation (14) with $P_s(2n,\beta) \le 1$ and $\delta \le \varepsilon$ gives $P_\tau(4n,\beta) \ge (1-\varepsilon)^2(1-\delta) \ge (1-\varepsilon)^3$. *Move 4* (squaring in time). Squaring $\rho_{4n}(\beta)$ to $\rho_{4n}(2\beta)$ gives $P_\tau(4n,2\beta) \ge 1-\delta \ge 1-5\varepsilon^2$, proving Eq. (15).

Starting estimate and iteration.— The doubling estimate needs one starting scale where both defects are below $1/100$. Proposition 5.5 of Ref. [1] gives the $j=0$ starting estimates:

$$P_s(2304,784),\ P_\tau(4608,784) \ge 0.99. \tag{17}$$

Equation (17) cannot come from diagonalizing X_β, which acts on whole bond histories; Ref. [1] uses only traces of small rings and one variational bound. It shifts $H_L \to H_L + \kappa LI$, with $\kappa = 700741/500000$ chosen so that $-\kappa$ lies just above the energy per site of a matrix-product trial state ψ on $N=72$ sites; the factor $e^{-\kappa\beta L}$ in $Z_L(\beta)$ cancels from both purity ratios (9). Since $Z_N(\beta)$ is a sum of positive Boltzmann factors and the variational principle gives $E_0(N) \le \langle\psi|H_N|\psi\rangle/\langle\psi|\psi\rangle < -\kappa N$, the ground-state factor alone gives $Z_N(\beta) > 1$. Combined with eigenvalue bounds from the twisted traces $\mathrm{Tr}_v(X_\beta^n U_g)$ of rings of at most twelve sites, with a spin rotation U_g of the histories on one virtual bond, this forces a dominant λ_0; Appendices I–M of the companion certificate note draw these networks and six doublings from $(36, 49/4)$ to Eq. (17). After $j \ge 0$ doublings, set

$$n_j = 2304\,2^j, \qquad \beta_j = 784\,2^j, \qquad \varepsilon_j = \frac{20^{-2^j}}{5}. \tag{18}$$

Since $\varepsilon_0 = 1/100$ and $\varepsilon_{j+1} = 5\varepsilon_j^2$, Eq. (15) gives $P_s(n_j,\beta_j), P_\tau(2n_j,\beta_j) \ge 1-\varepsilon_j$ by induction on j: each inductive step runs Moves 1–4 once. The defect falls doubly exponentially in j, $\varepsilon_0 = 10^{-2}$ $\varepsilon_4 \approx 3\times 10^{-22}$ (log scale), and so exponentially in β_j, $\varepsilon_j = \frac{1}{5}\exp[-\beta_j(\log 20)/784]$, as needed in Eq. (2).

All even lengths.— These estimates hold at the doubled lengths n_j and $2n_j$. To cover every even $L \ge 2304$, choose j with $n_j \le L \le 2n_j$. Since the weights $p_i = |\lambda_i(\beta)|^n/Z_n(\beta) \in [0,1]$ sum to one, any power $r \ge 1$ gives $Z_{rn}(\beta)/Z_n(\beta)^r = \sum_i p_i^r \le \sum_i p_i = 1$; taking $r = L/n_j \ge 1$ at β_j and $r = 2n_j/L \ge 1$ at $2\beta_j$ yields

$$Z_L(\beta_j) \le Z_{n_j}(\beta_j)^{L/n_j}, \qquad Z_L(2\beta_j) \ge Z_{2n_j}(2\beta_j)^{L/(2n_j)}. \tag{19}$$

Dividing $Z_L(2\beta_j)$ by $Z_L(\beta_j)^2$, substituting the torus identity $Z_{2n_j}(2\beta_j)/Z_{n_j}(\beta_j)^4 = P_\tau(2n_j,\beta_j)P_s(n_j,\beta_j)^2$ [Eq. (12)], and using $L/(2n_j) \le 1$ with $(1-\varepsilon_j)^3 \ge 1-3\varepsilon_j > \frac{1}{2}$ gives

$$P_\tau(L,\beta_j) \ge \big[P_\tau(2n_j,\beta_j)P_s(n_j,\beta_j)^2\big]^{L/(2n_j)} \ge 1-3\varepsilon_j. \tag{20}$$

Recovering the gap.— With $w_0 \ge w_1 \ge \cdots$ for $\rho_L(\beta_j)$, Eqs. (3) and (20) give $w_0 \ge 1-3\varepsilon_j > 1/2$: the Gibbs

spectrum looks like  rather than , so the ground state is unique and $w_1 \le 1 - w_0 \le 3\varepsilon_j$. Since $3\varepsilon_j/(1-3\varepsilon_j) \le (60/97)20^{-2^j} < 20^{-2^j}$,

$$e^{-\beta_j \Delta_L} = \frac{w_1}{w_0} \le \frac{1-w_0}{w_0} \le \frac{3\varepsilon_j}{1-3\varepsilon_j} < 20^{-2^j}. \qquad (21)$$

Taking logarithms and dividing by $\beta_j = 784\,2^j$ gives, for every even $L \ge 2304$,

$$\boxed{\Delta_L > \frac{\log 20}{784} \simeq 0.003821.} \qquad (22)$$

This proves Theorem 1, conditional on Eq. (17); in particular $\liminf_{L\to\infty,\, L \text{ even}} \Delta_L \ge (\log 20)/784 > 0$. The scale-doubling recurrence and the gap deduction are formally verified in Lean 4 (`lean/CertifiedGaps/PurityCriterion.lean`).

DISCUSSION: REPRESENTATION IN MACHINE-ASSISTED PROOFS

The uniform spectral gap theorems for the periodic and boundary-field spin-1 Heisenberg chains, together with the certified starting estimates, come from the AI-generated proofs by a new internal model of OpenAI [1, 2]. Because such AI-generated proofs tend to produce convoluted write-ups, we advocate tensor networks as a diagrammatic language that current AI models lack, to produce clearer understanding: here, the authors asked GPT-6 to re-examine the argument in terms of spacetime tensor networks and worked through the resulting structure interactively. Treating imaginary time and space alike on a torus reduces the self-adjoint spatial transfer operator and the contractive scale-doubling iteration to short identities between planar diagrams. Tensor-network diagrams are thus useful not only for calculation, but also for reading, simplifying, and extracting the physical mechanism of a machine-generated proof. The authors checked all mathematical and formal proofs.

[1] OpenAI, *The periodic spin-one Haldane gap*, manuscript dated September 24, 2026, Proposition 5.5 and Appendices A–C.

[2] OpenAI, *A boundary-field gap for the spin-one Heisenberg chain*, manuscript dated September 24, 2026, Secs. 2–6 and Appendices A and B.

[3] W. Tang, H.-H. Tu, and L. Wang, Continuous matrix product operator approach to finite temperature quantum states, Phys. Rev. Lett. **125**, 170604 (2020).

[4] B. Pirvu, V. Murg, J. I. Cirac, and F. Verstraete, Matrix product operator representations, New J. Phys. **12**, 025012 (2010).

[5] Companion note, *Scope of the spacetime bootstrap* (`thermal-mpo-gap-scope.tex`, 2026), Appendix E.

[6] T. Kennedy, Exact diagonalisations of open spin-1 chains, J. Phys.: Condens. Matter **2**, 5737 (1990).

Appendix A: Open chain with boundary fields

The tensor networks of the main text also bound the gap of the open chain with equal endpoint fields. On n spin-1 sites, with $\sum_\alpha (S^\alpha)^2 = 2I$, let

$$H_n^{\mathrm{s}} = \sum_{i=1}^{n-1} \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} - h(S_1^z + S_n^z), \qquad h = \frac{3}{5}. \tag{A1}$$

Theorem 2 (Boundary-field gap [2])**.** *For every odd $n \geq 1921$, the Hamiltonian (A1) has a unique ground state and $\Delta_n > (\log 10)/392 \simeq 0.005874$.*

The theorem, the field $h = 3/5$, and all finite-scale inputs in Eq. (C1) come from Ref. [2]. Appendices A–C prove the theorem from those inputs in the diagrammatic language of the main text.

Why are boundary fields needed? Without them, each end of the open spin-1 chain carries an effective spin-1/2 boundary mode, and the four lowest states are split only by an amount exponentially small in n [6]. Equal fields lift this boundary degeneracy on odd chains [2].

What must be shown? As on the ring, a thermal purity close to one forces a gap. With w_m the Gibbs weights of H_n^{s}, Eqs. (2) and (3) reduce Theorem 2 to a lower bound on the equal-field thermal purity $P_\tau^{\mathrm{s}}(n,\beta) = \sum_m w_m^2$:

$$e^{-\beta\Delta_n} = \frac{w_1}{w_0} \leq \frac{1 - P_\tau^{\mathrm{s}}(n,\beta)}{P_\tau^{\mathrm{s}}(n,\beta)}. \tag{A2}$$

We bound this purity along a sequence of scales where length and inverse temperature double together,

$$\beta_j = 49\,2^{j-2}, \quad n_j = 120\,2^{j-2}, \quad M_j = 2n_j \quad (j \geq 2), \tag{A3}$$

starting from the input scale $(n_2, \beta_2) = (120, 49)$ of Ref. [2]. Here n_j is a ring length and M_j is an open-chain length; each chain of length n with $M_j \leq n \leq 2M_j$ is handled at scale j. At every scale $j \geq 5$, we bound three defects by a single quantity x_j that squares (times ten) from one scale to the next:

$$\begin{aligned} &\varepsilon_j,\ e_{k_j}(j),\ 1 - P_\tau^{\mathrm{o}}(M_j,\beta_j) \leq x_j \quad (j \geq 5), \\ &x_5 = .01, \qquad x_{j+1} = 10x_j^2, \qquad x_j = 10^{-1-2^{j-5}}. \end{aligned} \tag{A4}$$

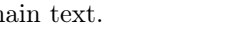
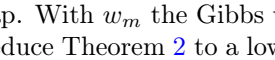
In words, the three claims are:

(R) *The ring is close to one eigenvalue:* the ring purity defect ε_j is at most x_j, so the transfer operator X_{β_j} has one dominant eigenvalue (Appendix B).

(E) *The ends see that eigenvector:* the boundary defect $e_{k_j}(j)$ of Eq. (A13), the fraction of the open-chain partition function not carried by the leading eigenvector, is at most x_j.

(T) *The open chain is close to its ground state:* the thermal defect $1 - P_\tau^{\mathrm{o}}(M_j,\beta_j)$ of an auxiliary open chain with opposite end fields [Eq. (A16)] is at most x_j.

Together, claims (R), (E), and (T) give $P_\tau^{\mathrm{s}}(n,\beta_j) \geq 1 - 4x_j$ for every $M_j \leq n \leq 2M_j$ [Eq. (A26)]. Since $x_j \leq .01$, the right-hand side of Eq. (A2) is at most $4x_j/(1-4x_j)$, which is less than $10^{-2^{j-5}}$, and taking logarithms with $\beta_j = 392\,2^{j-5}$ gives $\Delta_n > (\log 10)/392$. The three claims depend on each other as follows. Claim (R) concerns the ring alone, and Appendix B proves it from the ring inputs by the doubling estimate of the main text. In the induction step below, claims (E) and (T) at scale $j+1$ follow from (E) and (T) at scale j together with (R), while Appendix C supplies the base case $j = 5$ from the inputs at $j = 2$. The intervals $[M_j, 2M_j]$ with $j \geq 5$ cover every $n \geq M_5 = 1920$, which is where the bound $n \geq 1921$ (odd) comes from.

Why can the ring proof not be applied directly? An open chain  does not close into a ring, so its partition function is not a trace of X_β^n and there is no spatial purity to square. Its bulk still uses the transfer operator X_β, while boundary vectors cap the outer virtual legs of the first and last sites. In the classical open Ising chain, for example, the partition function is $u^{\mathsf{T}} K^{n-1} u$ with transfer matrix K and boundary vector $u = (e^{\beta h}, e^{-\beta h})$ for a field h on each end spin. Whereas a periodic trace  weights every eigenvector of X_β by one, an open chain weights the leading eigenvalue $\lambda(\beta)$ by the squared boundary overlap $c(\beta)$ [Eq. (A10)], which vanishes if the boundary vector is orthogonal to the leading eigenvector. For long chains, the open partition function is approximately $c(\beta)\lambda(\beta)^n$ and its thermal purity is approximately $[\lambda(2\beta)/\lambda(\beta)^2]^n c(2\beta)/c(\beta)^2$. The first factor is a bulk ratio controlled by the ring [claim (R)], while the second is a boundary ratio independent of n that cancels when we compare lengths M and $2M$. Doubling the length therefore costs only the bulk factor, and doubling the inverse temperature by squaring (10) needs no ring at all. This cancellation holds once the leading term $c(\beta)\lambda(\beta)^n$ dominates the remainder. Claim (E) states this through the boundary defect e, which shrinks as the chain grows.

What closes the ends of the chain? The open chain uses the bulk tensor W_β of Eq. (4), with the per-site energy shift $\kappa = 700741/500000$ of Ref. [2] (called a there) absorbed into each tensor as a factor $e^{-\kappa\beta}$, sending $H_n^\nu \to H_n^\nu + \kappa n I$. The shift places the leading periodic eigenvalue $\lambda_0(\beta_j)$ near one at the scales (A3) while canceling from every Gibbs state and purity. On an end site, the boundary field $\pm hS^z = \mp(ih)G_z$ acts like half a bond in the imaginary-time expansion, inserting G_z events of weight ih on the outer virtual leg. For a history a with $|a|$ events, the resulting boundary vector on $L^2(\mu_\beta)$ is

$$v_\beta(a) = \begin{cases} (ih)^{|a|}, & \text{all labels are } z, \\ 0, & \text{otherwise,} \end{cases} \qquad \|v_\beta\|^2 = e^{\beta h^2}, \tag{A5}$$

with value one on the empty history ($|a| = 0$). Integrating the ordered event times of the z labels gives $\|v_\beta\|^2$ as

the sum of $(\beta h^2)^{|a|}/|a|!$ over $|a| \geq 0$.

Which fields do the two caps produce? The conjugated left cap $\bar{v}_\beta$ weights each z event by $(-ih)G_z = +hS^z$ and adds $-hS_1^z$ to the Hamiltonian, whereas the unbarred right cap v_β weights each z event by $(ih)G_z = -hS^z$ and adds $+hS_n^z$. Capping both ends with the plain vector therefore gives the opposite-field chain $H_n^{\mathrm{o}} := H_n^{\mathrm{s}} + 2hS_n^z$, ↑•••••↓. To restore equal fields, we reverse the right field with the π rotation $P = \mathrm{diag}(1,-1,-1)$ about x, which flips $S^z \to -S^z$ and negates every z label, so its virtual action U_P on the right cap produces equal fields, ↑•••••↑. Opening the ring of Eq. (6) and capping its ends gives

$$e^{-\beta H_n^{\mathrm{s}}} = \bar{v}_\beta \text{—} W \text{—} W \text{—} \cdots \text{—} W \text{—} W \text{—} U_P v_\beta , \tag{A6}$$

and tracing the physical legs onto the transfer chain X_β^n of Eq. (8) yields the open-chain counterparts of Z_n in Eq. (7):

$$Y_n(\beta) = \mathrm{Tr}\, e^{-\beta H_n^{\mathrm{o}}} = \bar{v}_\beta \text{—} X_\beta^n \text{—} v_\beta , \quad ↑•••••↓, \tag{A7}$$

$$Y_n^{\mathrm{s}}(\beta) = \mathrm{Tr}\, e^{-\beta H_n^{\mathrm{s}}} = \bar{v}_\beta \text{—} X_\beta^n \text{—} U_P v_\beta , \quad ↑•••••↑. \tag{A8}$$

We work with the opposite-field chain first because its two caps use the same boundary vector, making its even moments nonnegative, and then transfer the bounds to H_n^{s} at the end. That transfer relies on $[U_P, X_\beta] = 0$, which holds because simultaneous spin rotations of the histories act on $L^2(\mu_\beta)$ as real orthogonal operators commuting with X_β. Before the κ shift, the series of absolute values for the traced expansion sums to at most $3^n e^{\beta[3(n-1)+2h]} < \infty$, so Fubini's theorem justifies regrouping the time-ordered series into the site-by-site contractions (A7)–(A8).

What does claim (R) say about X_β? At successive inverse temperatures $\beta_{j+1} = 2\beta_j$, the ring estimates of Appendix B show that the leading eigenvalue $\lambda_j := \lambda_0(\beta_j)$ of Eq. (7) is simple and positive, ▮. Normalizing λ_j to one splits X_{β_j}/λ_j into its leading projector Π_j and an orthogonal remainder R_j ($\Pi_j R_j = R_j \Pi_j = 0$):

$$T_j := X_{\beta_j}/\lambda_j = \Pi_j + R_j, \qquad \Pi_j = |\psi_j\rangle\langle\psi_j|, \quad \|\psi_j\| = 1, \tag{A9}$$

so $T_j^m = \Pi_j + R_j^m$. Appendix B proves that $\|R_j\| < 1$, $U_P \Pi_j = \Pi_j$, and $0 < r_j := \lambda_{j+1}/\lambda_j^2 \leq 1$, with quantitative bounds in Eqs. (B1)–(B2).

How much of the boundary vector lies along the leading eigenvector? With $v_j = v_{\beta_j}$, inserting the leading projector between the two caps cuts the network into two pieces:

$$\underbrace{\bar{v}_j \text{—} \Pi_j \text{—} v_j}_{c_j} = \bar{v}_j \text{—} \psi_j \quad \bar{\psi}_j \text{—} v_j = |\langle \psi_j, v_j \rangle|^2. \tag{A10}$$

The overlap c_j is the squared projection of v_j onto ψ_j, ∠. Whereas on a ring the same insertion gives $\mathrm{Tr_v}\, \Pi_j = 1$, here it gives c_j, which vanishes when the two vectors are orthogonal. For even $m \geq 2$, substituting $T_j^m = \Pi_j + R_j^m$ splits the normalized partition function $B_m(j) := Y_m(\beta_j)/\lambda_j^m > 0$ into c_j and a remainder $D_m(j)$:

$$\underbrace{\bar{v}_j \text{—} T_j^m \text{—} v_j}_{B_m(j)} = \underbrace{\bar{v}_j \text{—} \Pi_j \text{—} v_j}_{c_j} + \underbrace{\bar{v}_j \text{—} R_j^m \text{—} v_j}_{D_m(j)} . \tag{A11}$$

Because R_j is self-adjoint, factoring $R_j^m = R_j^{m/2} R_j^{m/2}$ and applying one half to each boundary vector folds the remainder into a nonnegative squared norm:

$$D_m(j) = \begin{array}{l} R_j^{m/2} \text{—} v_j \\ R_j^{m/2} \text{—} \bar{v}_j \end{array} = \|R_j^{m/2} v_j\|^2 \geq 0. \tag{A12}$$

The *boundary defect* of claim (E) is the remainder's share of the partition function,

$$e_m(j) := \frac{D_m(j)}{B_m(j)} = 1 - \frac{c_j}{B_m(j)}, \tag{A13}$$

shown as the grey fraction of ▮ at the starting value $e_{142}(2) \leq .14$ of Eq. (C1). Finding $e_m < 1$ guarantees a nonzero overlap with the leading eigenvector.

Why does a longer chain help? Extending the chain shrinks the remainder while leaving c_j unchanged. For even $m \geq k$, move $k/2$ factors of R_j onto each boundary vector and set $y = R_j^{k/2} v_j$:

$$D_m(j) = \bar{y} \text{—} R_j^{m-k} \text{—} y \leq \|R_j\|^{m-k} \underbrace{\|y\|^2}_{D_k(j)} . \tag{A14}$$

The remainder therefore decays geometrically with the extra length $m-k$, ⌒, leaving $c_j \leq B_m(j) \leq B_k(j)$.

How are two temperatures connected? Equation (A6) is an exact operator identity on the physical Hilbert space, so composing $e^{-\beta H_m^{\mathrm{o}}} e^{-\beta H_m^{\mathrm{o}}} = e^{-2\beta H_m^{\mathrm{o}}}$ in imaginary time links β to 2β:

$$\underbrace{\begin{array}{l} W_\beta \text{—} W_\beta \text{—} \cdots \text{—} W_\beta \\ W_\beta \text{—} W_\beta \text{—} \cdots \text{—} W_\beta \end{array}}_{e^{-\beta H_m^{\mathrm{o}}}\, e^{-\beta H_m^{\mathrm{o}}}} = \underbrace{W_{2\beta} \text{—} W_{2\beta} \text{—} \cdots \text{—} W_{2\beta}}_{e^{-2\beta H_m^{\mathrm{o}}}} . \tag{A15}$$

This equality holds only for the full operator product, as two W_β tensors do not fuse into one $W_{2\beta}$ site by site. Normalizing the trace defines the opposite-field Gibbs state $\rho_m^{\mathrm{o}}(\beta) = e^{-\beta H_m^{\mathrm{o}}}/Y_m(\beta)$, whose thermal purity is the quantity in claim (T):

$$P_\tau^{\mathrm{o}}(m, \beta_j) := \frac{Y_m(\beta_{j+1})}{Y_m(\beta_j)^2} = \mathrm{Tr}\, \rho_m^{\mathrm{o}}(\beta_j)^2 \in (0, 1]. \tag{A16}$$

Substituting $Y_m(\beta_j) = \lambda_j^m B_m(j)$ and $\lambda_{j+1} = r_j \lambda_j^2$ into Eq. (A16) links the normalized partition functions at adjacent temperatures:

$$B_m(j+1) = P_\tau^{\mathrm{o}}(m, \beta_j) r_j^{-m} B_m(j)^2. \tag{A17}$$

(E) Why does the boundary defect shrink from one scale to the next? In Move 1, we advance the boundary defect from scale j to scale $j+1$ (marked by primes).

Assume $e_k(j) < 1$, choose even $m > k \geq 2$, and let $\|R_{j+1}\|^{m-k} \leq q < 1$. The suppression bound (A14) at scale $j+1$ gives $D'_m \leq qD'_k$. Since c' does not depend on the chain length, $D'_k - D'_m = B'_k - B'_m$. Substituting $D'_k = D'_m + B'_k - B'_m$ into $D'_m \leq qD'_k$ yields the first inequality below, and dividing by B'_m gives the second:

$$(1-q)D'_m \leq q(B'_k - B'_m), \qquad e'_m \leq \frac{q}{1-q}\left(\frac{B'_k}{B'_m} - 1\right). \tag{A18}$$

To bound the unknown ratio B'_k/B'_m at the new temperature, we rewrite it at the old temperature with Eq. (A17) and apply $P^{\mathrm{o}}_\tau(k,\beta_j), r_j \leq 1$ together with $B_m(j) \geq c_j = [1-e_k(j)]B_k(j) > 0$:

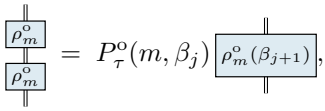
$$\begin{aligned}\frac{B'_k}{B'_m} &= \frac{P^{\mathrm{o}}_\tau(k,\beta_j)}{P^{\mathrm{o}}_\tau(m,\beta_j)}\left[\frac{B_k(j)}{B_m(j)}\right]^2 r_j^{m-k} \\ &\leq \frac{1}{P^{\mathrm{o}}_\tau(m,\beta_j)[1-e_k(j)]^2}.\end{aligned} \tag{A19}$$

Because the denominator uses only the old overlap, finding $e'_m < 1$ certifies that the new overlap is also positive. If $e_k(j)$ and $1 - P^{\mathrm{o}}_\tau(m,\beta_j)$ are $\mathcal{O}(x)$, then the right-hand side of Eq. (A19) is $1 + \mathcal{O}(x)$, and choosing $m-k$ large enough that $q \leq \mathcal{O}(x)$ makes e'_m of order x^2.

(T) Why does doubling the length keep the purity? Once Move 1 controls the new boundary defect, Move 2 doubles the chain length at almost no cost in purity. Solve Eq. (A17) for $P^{\mathrm{o}}_\tau(m,\beta_j)$, divide the $m = 2M$ relation by the $m = M$ relation for even M, and bound the two partition-function ratios using $B_{2M}(j+1) \geq c_{j+1} = [1-e_M(j+1)]B_M(j+1)$ and $B_M(j) \geq B_{2M}(j)$ [Eq. (A14)]:

$$\frac{P^{\mathrm{o}}_\tau(2M,\beta_j)}{P^{\mathrm{o}}_\tau(M,\beta_j)} = r_j^M \underbrace{\frac{B_{2M}(j+1)}{B_M(j+1)}}_{\geq 1-e_M(j+1)} \underbrace{\left[\frac{B_M(j)}{B_{2M}(j)}\right]^2}_{\geq 1}. \tag{A20}$$

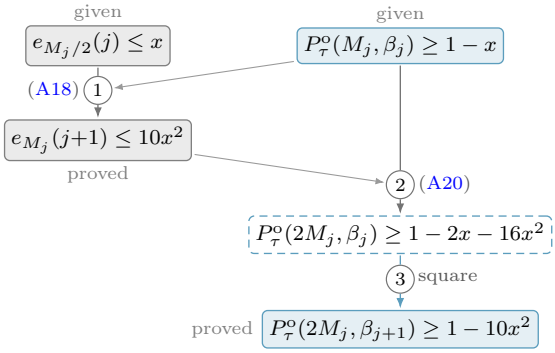
Because the bulk factor r_j^M is close to one by Eq. (B2), doubling the length lowers the purity only slightly. Move 3 then doubles the inverse temperature by squaring the Gibbs operator, as in Eq. (10):



(A21)

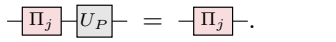
and applying the squaring bound (11) at $m = 2M$ brings $1 - P^{\mathrm{o}}_\tau$ down to order x^2.

How do the three moves make an induction? Each step $j \to j+1$ doubles both length and inverse temperature, using boundary-defect lengths $k_2 = 142$ (matching the input $e_{142}(2)$) and $k_{j+1} = M_j$. For $j \geq 3$, this choice sets $k_j = M_j/2$, leaving $M_j/2$ sites to suppress the remainder in Move 1. The ring defect ε_j ($\varepsilon_2 = .052$, $\varepsilon_{j+1} = 6\varepsilon_j^2$) is bounded in Appendix B [Eq. (B1)]. Starting from the inputs at $j = 2$, three explicit steps in Appendix C reach $j = 5$, where induction on $j \geq 5$ takes over to prove Eq. (A4). Read the diagram below from top to bottom, with grey boxes for the boundary defect $e_k(j)$ [claim (E)] and blue boxes for the opposite-field thermal purity P^{o}_τ [claim (T)]. Solid boxes are given at scale j or proved at scale $j+1$, the dashed box is intermediate, and each grey arrow supplies the second input to a move:



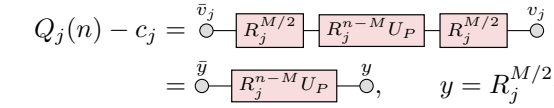
Here Move 1 is the overlap recurrence (A18)–(A19), Move 2 is the length doubling (A20), and Move 3 is the squaring (A21). Since $M_{j+1} = 2M_j$, the proved pair becomes the given pair at scale $j+1$, so the step repeats. The expressions $10x^2$ and $1-2x-16x^2$ in the boxes are verified at the end of Appendix C. The defect bound falls doubly exponentially, $x_5 = 10^{-2} \to x_9 = 10^{-17}$ (log scale), and hence exponentially in $\beta_j = 392\,2^{j-5}$.

How do the bounds pass to equal fields and odd lengths? Claims (E) and (T) control the opposite-field chain at even lengths M_j, whereas the theorem requires the equal-field chain at odd n. The rotation U_P links the two partition functions [Eq. (A8)] and leaves the leading projector unchanged:



(A22)

The normalized equal-field partition function $Q_j(n) := Y^{\mathrm{s}}_n(\beta_j)/\lambda_j^n$ therefore has the same leading term c_j. For every even $M \leq n$, commuting U_P through the right-hand factors ($[U_P, R_j] = 0$) isolates the remainder:



(A23)

Because $\|R_j^{n-M}U_P\| \leq 1$,

$$|Q_j(n) - c_j| \leq \|y\|^2 = D_M(j), \tag{A24}$$

which bounds the remainder at odd length by $D_M(j)$ without needing R_j^n to be positive.

What equal-field purity follows? For $M_j \leq n \leq 2M_j$ and $M = M_j = k_{j+1}$, Eq. (A24) gives $Q \leq c + D_M = B_M$ and $Q \geq c - D_M = (1-2e_M)B_M$. With $e_M(j+1) \leq x_{j+1}$ from claim (E) [Eq. (A4)],

$$Q_j(n) \leq B_M(j), \quad Q_{j+1}(n) \geq (1-2x_{j+1})B_M(j+1) > 0. \tag{A25}$$

Write $P^{\mathrm{s}}_\tau(n,\beta_j) = Q_{j+1}(n)\lambda_{j+1}^n/[Q_j(n)\lambda_j^n]^2$ and substitute these bounds together with Eq. (A17) at $m = M$.

Using $r_j^{n-M} \geq r_j^M$ [Eq. (B2)] and claims (R) and (T) ($\varepsilon_j \leq x_j$, $\varepsilon_{j+1} \leq 6x_j^2$, and $x_{j+1} = 10x_j^2$) then gives

$$\begin{aligned} P_\tau^{\mathrm{s}}(n,\beta_j) = \frac{Y_n^{\mathrm{s}}(\beta_{j+1})}{Y_n^{\mathrm{s}}(\beta_j)^2} &\geq (1-2x_{j+1})P_\tau^{\mathrm{o}}(M,\beta_j) r_j^{n-M} \\ &\geq (1-2x_{j+1})(1-x_j)(1-\varepsilon_j)(1-\varepsilon_{j+1}) \\ &\geq 1 - 2x_j - 26x_j^2 \geq 1 - 4x_j. \end{aligned} \tag{A26}$$

Proof of Theorem 2. The intervals $[M_j, 2M_j]$ cover every integer $n \geq M_5 = 1920$. For odd $n \geq 1921$, choose $j \geq 5$ with $M_j \leq n \leq 2M_j$. The largest Gibbs weight satisfies $w_0 \geq P_\tau^{\mathrm{s}}(n,\beta_j) \geq 1 - 4x_j > 1/2$, giving a spectrum like ▙ with a unique ground state. Equation (A2) gives

$$e^{-\beta_j \Delta_n} \leq \frac{4x_j}{1-4x_j} \leq \frac{5}{12}\, 10^{-2^{j-5}} < 10^{-2^{j-5}}. \tag{A27}$$

Taking logarithms and using $\beta_j = 392\, 2^{j-5}$ yields $\Delta_n > (\log 10)/392$. □

Appendix B: Transfer bounds at the open-chain scales

We now prove claim (R) of Appendix A by showing that the ring purity defect ε_j at the scales (A3) starts at $\varepsilon_2 = .052$ and shrinks by squaring (times six) at each step. From this purity bound we derive the transfer-operator properties used in Appendix A, namely the remainder bound

$$\begin{aligned} \|R_j\|^{n_j} \leq d(\varepsilon_j), \qquad d(\varepsilon) = \frac{\varepsilon}{2-3\varepsilon}, \\ \varepsilon_2 = .052, \qquad \varepsilon_{j+1} = 6\varepsilon_j^2, \end{aligned} \tag{B1}$$

and the eigenvalue and symmetry relations

$$0 < r_j \leq 1, \quad r_j^{M_j} \geq (1-\varepsilon_j)(1-\varepsilon_{j+1}), \quad U_P \Pi_j = \Pi_j. \tag{B2}$$

Why does the ring defect square at each scale? The bound follows from the first row of Eq. (C1) by the doubling estimate of the main text. Because the starting defect $\varepsilon_2 = 1 - .948 = .052$ exceeds the threshold $1/100$ of Eq. (15), the prefactor 5 there becomes 6. If $P_{\mathrm{s}}(n,\beta), P_\tau(2n,\beta) \geq 1-\varepsilon$, then Move 1 of Eq. (15) gives $P_{\mathrm{s}}(n,2\beta) \geq (1-\varepsilon)^3$, and squaring with Eq. (11) bounds $1 - P_{\mathrm{s}}(2n,2\beta)$ by

$$f(1-(1-\varepsilon)^3) = \frac{\varepsilon^2}{2}\left[\sum_{s=1}^{3}(1-\varepsilon)^{-s}\right]^2 \leq 6\varepsilon^2 \tag{B3}$$

for $0 \leq \varepsilon \leq .052$, in place of $5\varepsilon^2$. Since $6\varepsilon^2 \leq \varepsilon$, Moves 3 and 4 give the same bound for $1 - P_\tau(4n,2\beta)$. Induction therefore proves $P_{\mathrm{s}}(n_j,\beta_j), P_\tau(M_j,\beta_j) \geq 1 - \varepsilon_j$.

How does a purity bound control the remainder? A purity close to one forces one probability to dominate, ▙, and the rest to be small. For a probability distribution with purity at least $1 - \varepsilon > 1/2$ and largest weight p_0, set $z = 1 - p_0$. Because the purity cannot exceed p_0, we have $p_0 > 1/2$, making the largest weight unique and $z < 1/2$. The purity is also at most $p_0^2 + z^2 = 1 - 2z(1-z)$, so $2z(1-z) \leq \varepsilon$. Combining this with $z < 1/2$ gives $z \leq 2z(1-z) \leq \varepsilon$, which implies $z \leq \varepsilon/[2(1-\varepsilon)]$ and $z/(1-z) \leq d(\varepsilon)$. Applying this inequality to the eigenvalues $p_i = |\lambda_i(\beta_j)|^{n_j}/Z_{n_j}(\beta_j)$ of $\sigma_{n_j}(\beta_j)$ shows that the largest-modulus eigenvalue is simple and satisfies

$$\sum_{i \geq 1} \left|\frac{\lambda_i(\beta_j)}{\lambda_j}\right|^{n_j} \leq d(\varepsilon_j), \tag{B4}$$

which implies Eq. (B1). Finally, the leading eigenvalue is positive, since $Z_N(\beta_j) > 0$ for all large odd N whereas a negative leading eigenvalue would make $\mathrm{Tr}_{\mathrm{v}}\, X_{\beta_j}^N$ negative once the subleading sum decays (as X_{β_j} is Hilbert–Schmidt with remaining spectral radius strictly smaller than λ_j).

Why is the leading eigenvector unchanged by U_P? The relation $U_P \Pi_j = \Pi_j$ in Eq. (B2) allows Appendix A to pass from opposite to equal fields. The real transfer kernel admits real orthogonal virtual actions of the three π rotations and of the cyclic permutation of the coordinate axes, all commuting with X_{β_j}. Each of these symmetries acts as a sign on the real one-dimensional leading eigenspace. Because the order-three cyclic permutation has sign $+1$ and conjugates the three π rotations into one another, all three π rotations share the same sign; since the product of two is the third, that common sign is $+1$, giving $U_P \Pi_j = \Pi_j$.

How close is r_j to one? The ratio $r_j = \lambda_{j+1}/\lambda_j^2$ is the bulk factor of the length doubling in Appendix A [Eq. (A20)]. Its upper bound holds because the thermal purity $P_\tau(N,\beta_j)$ is at most one, so $Z_N(\beta_{j+1}) \leq Z_N(\beta_j)^2$, and taking Nth roots along even $N \to \infty$ gives $\lambda_{j+1} \leq \lambda_j^2$. For the lower bound at the finite length $M_j = n_{j+1}$, where the largest eigenvalue of $\sigma_{M_j}(\beta_{j+1})$ is at least $1 - \varepsilon_{j+1}$, combining this spatial weight bound with the thermal purity bound at scale j gives

$$\begin{aligned} \lambda_{j+1}^{M_j} &\geq (1-\varepsilon_{j+1}) Z_{M_j}(\beta_{j+1}) \\ &\geq (1-\varepsilon_{j+1})(1-\varepsilon_j) Z_{M_j}(\beta_j)^2 \\ &\geq (1-\varepsilon_{j+1})(1-\varepsilon_j)\lambda_j^{2M_j}, \end{aligned}$$

which is the remaining bound in Eq. (B2).

Appendix C: Starting estimates for the open chain

We now establish the base case of the induction in Appendix A: claims (E) and (T) at $j = 5$, with all three defects of Eq. (A4) below $x_5 = .01$. Because the certified inputs of Ref. [2] at $j = 2$ have open-chain defects .14 and .21, above the threshold $x \leq .01$ of the general induction step, we bridge $j = 2$ to $j = 5$ in three explicit steps. Below we state the inputs, write the three moves of Appendix A as numerical update rules, run those rules for three steps, and verify the induction step for $j \geq 5$.

What are the inputs? From periodic variational estimates and small-cluster boundary computations, Ref. [2] certifies

$$\begin{aligned} P_{\mathrm{s}}(120,49) \ge .948, \qquad & P_\tau(240,49) \ge .948, \\ e_{142}(2) \le .14, \qquad & P_\tau^{\mathrm{o}}(240,\beta_2) \ge .79. \end{aligned} \tag{C1}$$

The first row gives the ring purity defect $\varepsilon_2 = 1 - .948 = .052$ at $(n_2, \beta_2) = (120, 49)$, used for claim (R) in Appendix B. The second row gives the open-chain boundary defect at $k_2 = 142$ [claim (E)] and the opposite-field thermal purity at $M_2 = 240$ [claim (T)].

How do the three moves become numbers? Write E_j for an upper bound on $e_{k_j}(j)$ and η_j for an upper bound on $1 - P_\tau^{\mathrm{o}}(M_j, \beta_j)$. In Move 1, combining Eqs. (A18) and (A19) bounds the new boundary defect by

$$\begin{aligned} e_m(j+1) &\le F(q, e_k(j), P_\tau^{\mathrm{o}}(m, \beta_j)), \\ F(q,e,t) &= \frac{q}{1-q}\left[\frac{1}{(1-e)^2 t} - 1\right], \end{aligned} \tag{C2}$$

which increases with q, e and decreases with t. The suppression factor q must bound $\|R_{j+1}\|^{M_j - k_j}$. Because Eq. (B1) bounds $\|R_{j+1}\|^{n_{j+1}}$ and $n_{j+1} = M_j$, taking the $(M_j - k_j)/M_j$ power fixes the exponent in Eq. (C3). Moves 2 and 3 control the purity through Eqs. (A20) and (11). With $k_2 = 142$ and $k_{j+1} = M_j$, the updates are

$$q_j = d(\varepsilon_{j+1})^{(M_j - k_j)/M_j}, \tag{C3}$$

$$E_{j+1} \ge F(q_j, E_j, 1 - \eta_j), \tag{C4}$$

$$\eta_{j+1} \ge f\big(1 - (1-\eta_j)(1-\varepsilon_j)(1-\varepsilon_{j+1})(1-E_{j+1})\big), \tag{C5}$$

for every pair $E_{j+1}, \eta_{j+1} \in [0, 1)$ satisfying these inequalities.

What do three steps give? Starting from $\varepsilon_2 = .052$, $E_2 = .14$, and $\eta_2 = 1 - .79 = .21$, three applications of Eqs. (C4)–(C5) give the rational upper bounds:

j	E_j	η_j	q_j (upper bound)
2	.14	.21	.142
3	.12	.155	.0282
4	.016	.026	.0028
5	.001	.001	—

Both defects collapse along the iteration, with E_j falling as  and η_j as  ($j = 2, \ldots, 5$, each relative to $j = 2$). The first step gains little because $q_2 \le .142$ and the initial defects are still large, whereas once q_j is small the defects square. Rational arithmetic confirms these step bounds: at $j = 2$, for example, the exponent in Eq. (C3) is $(240 - 142)/240 = 49/120$, so $q_2 \le .142$ amounts to $d(\varepsilon_3)^{49} < .142^{120}$, and the later steps are verified by squaring. Together with $\varepsilon_5 < .000015$, these bounds establish the base case $j = 5$ of Eq. (A4).

Why does the induction step close? Assume Eq. (A4) at scale $j \ge 5$ with $x = x_j \le .01$. The relation $k_j = M_j/2$ gives

$$\begin{aligned} q_j &\le \sqrt{\frac{6x^2}{2 - 18x^2}} \le 2x, \\ F(2x, x, 1-x) &\le \frac{6x^2}{(1-2x)(1-x)^3} < 7x^2. \end{aligned} \tag{C6}$$

The choice $E_{j+1} = 10x^2$ therefore satisfies Eq. (C4). With $\varepsilon_{j+1} \le 6x^2$, the four-factor product in Eq. (C5) is at least $1 - 2x - 16x^2$, and $f(2x + 16x^2) < 3x^2$, so $\eta_{j+1} = 10x^2$ satisfies Eq. (C5). Because the prefactor 10 in $x_{j+1} = 10x_j^2$ covers all three bounds $7x^2$, $3x^2$, and $6x^2$, the induction for Eq. (A4) is complete.